PROJECT 09 / RESEARCH SOFTWARE

Nuclear Isotope Identification Lab

Scientific verification report | v1.0.0

Recorded verification: 2026-10-15T08:59:46.640Z

Implementation and DOM checks passed. Real-browser verification remains pending.

This report records what was actually executed. It does not certify isotope-identification accuracy, a false-positive rate, or a calibrated confidence probability. The example measurements and analytical fixtures are explicitly simulated.

Recorded metric	Result
Production reference library	11 parent states / 667 nuclear gamma transitions
Executed Vitest checks	115 passed / 0 failed in 8 files
Test breakdown	81 scientific/data-identity + 21 parser + 13 React DOM
Independent function audit	10 analytical cases passed
Actual browser tests	0 executed / 18 configured cases NOT RUN
Independent experimental campaign	Not performed
GitHub / Pages publication	Not performed

Use boundary

Research-oriented gamma-ray isotope identification support. Candidate identifications are not definitive without suitable measurement quality, calibrated detector response, validated methods and expert review. A single gamma line is never treated as conclusive proof.

Executed technical checks

Exit codes below are taken from the retained command records. Listing Playwright tests is not executing them.

Command actually executed	Exit
npm ci	0
npm run validate:data	0
python3 scripts/import_iaea.py	0
npm test	0
npm run typecheck	0
npm run build	0
node scripts/check-standalone.mjs	0
npx playwright test --list	0

Reference integrity and reproducibility

All 667 normalized records were checked against retained raw IAEA source rows and SHA-256 hashes. The offline importer reproduced the normalized JSON, CSV and manifest byte for byte. Source extraction and nuclear evaluation dates remain separate.

The production standalone HTML was executed in jsdom with an opaque file origin, including initial scientific results, SVG DOM, Arabic RTL, English and analysis interaction. This verifies bundle execution and handlers; it does not verify real visual layout, Canvas rasterization or browser downloads.

Browser checks not executed

Nine scenarios are configured for each of desktop Chromium and mobile Chromium: simulation/chart, CSV upload, manual input, tool navigation, language/theme, PDF download, CSV download, PNG download, and reference/responsive checks. The supported browser-control capability was unavailable in the preparation environment. These 18 cases are delivered for later actual execution.

Ten-function result summary

Reference expectations are independently stated analytical results. Full input, method, source and limit details follow.

ID	Function	Reference and observed result	State
01	Energy line matching	delta 0.6; sigma sqrt(0.5); z 0.848528	PASS
02	Candidate ranking	2 independent + 2 exclusive; 28 points	PASS
03	Multi-line consistency	Correlated centroid chi-square = 2/3	PASS
04	Missing expected lines	1000 predicted counts; conditional missing flag	PASS
05	Shared-line ambiguity	One shared group; two parent explanations	PASS
06	Relative intensity	Observed ratio 1; detector-weighted ratio 1; z=0	PASS
07	Mixture attribution	Two-component explanation; coverage 2/3	PASS
08	Evidence coverage	Peak 0.5; weighted 0.75; listed-area 0.1	PASS
09	Alternative hypotheses	Matches analytical Gaussian-plus-uniform log density	PASS
10	Compound evidence report	Single-line candidate remains Tentative	PASS

The detailed Markdown matrix in docs/FINAL_VALIDATION_REPORT.md contains all requested columns: function, scientific question, inputs, method, source, expected result, actual result, tests, pass/fail and use limits. The following pages carry the same review information as per-function tables.

01 Gamma energy matching

Review field	Verified content
Scientific question	Which line is energy compatible?
Inputs	Analytical fixture E=100.6, Eref=100, u=(0.3,0.4,0.5) keV
Method	Quadrature marginal uncertainty and signed normalized residual
Source / grounding	Analytical uncertainty fixture; IAEA for production photon data
Independent reference result	delta=0.6 keV; sigma=sqrt(0.5); z=0.6/sqrt(0.5)
Actual application result	delta=0.5999999999999943; sigma=0.7071067811865476; z=0.8485281374238489
Executed tests	src/physics/energyMatching.test.ts
Audit status	PASS
Use limits	Compatibility does not establish parent identity; missing uncertainty uses a declared non-statistical window.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

02 Candidate ranking

Review field	Verified content
Scientific question	Which parent has independent support?
Inputs	Two exact, exclusive independent analytical lines
Method	Maximum-cardinality assignment; S=10N+3U+2q
Source / grounding	Explicit project scoring model, not an IAEA probability
Independent reference result	N=2; U=2; evidence points=28
Actual application result	N=2; U=2; score=28
Executed tests	src/physics/evidence.test.ts
Audit status	PASS
Use limits	Uncalibrated ranking points; contradictions are separately assessed.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

03 Multi-line consistency

Review field	Verified content
Scientific question	Are correlated centroid residuals consistent?
Inputs	Residuals (1,1) keV; independent u=1; common ucal=1
Method	V=I+11^T; delta^T V^-1 delta
Source / grounding	Closed-form matrix inverse; JCGM covariance framework
Independent reference result	chi-square=2/3; degrees of freedom=2
Actual application result	chi-square=0.6666666666666667; dof=2
Executed tests	src/physics/evidence.test.ts
Audit status	PASS
Use limits	Gaussian energy diagnostic; no count chi-square, p-value or presence probability.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

04 Missing expected lines

Review field	Verified content
Scientific question	Should an unobserved line exceed the declared sensitivity threshold?
Inputs	Anchor N=1000; P=(0.5,0.25); efficiency=(0.1,0.2); threshold=20; complete list; assessed response
Method	Detector-weighted anchor ratio and lower expected-mean bound
Source / grounding	Analytical simulated response, explicitly declared; no experimental claim
Independent reference result	Expected mean 1000 counts; conditional expected-not-seen
Actual application result	expected=1000; state=expected-not-seen; lower=895.0428658927846
Executed tests	src/physics/response.test.ts
Audit status	PASS
Use limits	Requires complete sensitivity/uncertainty information and a pure anchor; not proof of absence.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

05 Shared-line ambiguity

Review field	Verified content
Scientific question	Does the observed group have competing parents?
Inputs	One 100.05 keV fixture peak; Sim-A=100, Sim-B=100.1
Method	Distinct compatible parent sets per independent group
Source / grounding	Analytical ambiguity fixture, not real nuclide records
Independent reference result	One shared group containing Sim-A and Sim-B
Actual application result	groups=1; parents=Sim-A, Sim-B
Executed tests	src/physics/evidence.test.ts
Audit status	PASS
Use limits	Discriminating lines still need detector-sensitivity assessment.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

06 Relative intensity consistency

Review field	Verified content
Scientific question	Do response-corrected ratios agree?
Inputs	Equal 1000-count net areas; P ratio=0.5; efficiency ratio=2
Method	Observed area ratio versus P*efficiency*s ratio; diagnostic log residual
Source / grounding	Analytical detector-response fixture
Independent reference result	Observed=1; expected=1; diagnostic z=0
Actual application result	observed=1; expected=1; z=0
Executed tests	src/physics/response.test.ts
Audit status	PASS
Use limits	No raw-area/probability comparison; low-count and unassessed-model residuals withheld.

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07 Mixed-isotope attribution

Review field	Verified content
Scientific question	Which bounded parent sets explain the independent groups?
Inputs	Separate Sim-A 200 and Sim-B 300 keV supports; unexplained 888 keV
Method	Enumerated cover objective with component penalty
Source / grounding	Analytical mixture fixture
Independent reference result	Best set Sim-A+Sim-B; coverage 2/3; unexplained u
Actual application result	set=Sim-A+Sim-B; coverage=0.6666666666666666; unexplained=u
Executed tests	src/physics/evidence.test.ts
Audit status	PASS
Use limits	Bounded search, no activity or concentration fractions.

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08 Evidence coverage

Review field	Verified content
Scientific question	How much listed evidence is explained?
Inputs	Two peaks; only first explained; weights (3,1); net areas (10,90)
Method	Explicit, distinct peak/group/weight/area denominators
Source / grounding	Independent hand-computed fractions
Independent reference result	Peak fraction=0.5; weighted=0.75; listed-area=0.1
Actual application result	peak=0.5; weighted=0.75; listed-area=0.1
Executed tests	src/physics/evidence.test.ts
Audit status	PASS
Use limits	Listed-area fraction excludes continuum and unlisted counts; none is a probability.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

09 Alternative hypotheses

Review field	Verified content
Scientific question	What does the specified centroid-density model predict?
Inputs	One exact 100 keV fixture centroid; u=0.1; q=0.9; range width 980 keV
Method	Gaussian-plus-uniform conditional density
Source / grounding	Closed-form likelihood independent of the implementation
Independent reference result	log[0.9/(sqrt(2*pi)*0.1)+0.1/980]=1.278314463549419
Actual application result	logL=1.278314463549419; relative=1
Executed tests	src/physics/hypothesisComparison.test.ts
Audit status	PASS
Use limits	One model is not a relative comparison; relative likelihood is never a posterior probability.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

10 Evidence report

Review field	Verified content
Scientific question	What conclusion is justified by the compound evidence?
Inputs	One exact, exclusive line; no quantitative detector model
Method	Conservative verbal synthesis and explicit limitations
Source / grounding	Declared project rules; validated analytical single-line boundary
Independent reference result	Tentative; independentExperimentalValidation=false; classesCalibrated=false
Actual application result	status=Tentative; experimental=false; calibrated=false
Executed tests	src/physics/identificationReport.test.ts
Audit status	PASS
Use limits	Uncalibrated research classification, not a definitive isotope certificate.

Analytical fixtures use explicitly simulated labels, not real nuclide identities. They are test-only records and are not present in the production nuclear gamma reference library.

Sources and remaining scientific limits

The production numerical source is IAEA LiveChart decay radiations derived from ENSDF. Its API guide describes an April 2022 ENSDF snapshot; source retrieval was 2026-10-14. Row-level evaluator names, cutoff dates, source URLs, parent excitation and original line numbers are retained.

Independent LNHB Co-60 photon-table spot checks agree with the selected photon energies and absolute yields. A newer LNHB Cs-137 evaluation differs slightly from the bundled older IAEA row; the difference is disclosed and not silently merged. This is a versioned reference subset, not a claim of latest ENSDF.

Calibration covariance is a shared offset only. Reference-energy/yield covariances are not supplied. Efficiency correlation is configurable; other independence assumptions and extra response-model uncertainty are stated. Missing-line contradictions require full sensitivity and response information.

The mixture cover is bounded and heuristic. Conditional centroid likelihoods depend on an explicit line-density and clutter model. Evidence points, verbal classes, coverage fractions and relative likelihoods are not calibrated isotope-presence probabilities.

Real-browser downloads, Canvas PNG, responsive/mobile/RTL visual layout and actual GitHub CI/Pages operation remain unexecuted. An independent labelled experimental benchmark is still needed before any performance or confidence claim.

Primary sources

https://www-nds.iaea.org/relnsd/vcharthtml/api_v0_guide.html

https://www-nds.iaea.org/relnsd/vcharthtml/guide.html

https://www.nndc.bnl.gov/ensdf/

https://www.lnhb.fr/nuclides/Co-60_tables.pdf

https://www.lnhb.fr/nuclides/Cs-137_tables.pdf

https://www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf

https://nucleus.iaea.org/Pages/Others/Terms-Of-Use.aspx